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Safety Features of Operating Light Water Reactors of Western Design

Mirela Gavrilas, Pavel Hejzlar, Neil E.
Todreas, Youssef Shatilla



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SAFETY FEATURES
of
OPERATING
LIGHT WATER REACTORS
of
WESTERN DESIGN

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The preparation of this manuscript involved considerable attention to editorial accuracy. Paula Cornelio oversaw this function as well as typing the manuscript and drawing most of the figures. James Dutton technically edited the manuscript and suggested numerous improvements for clarity and completeness. Thomas Ippolito participated in the early stages of gathering pressurized water reactor technical characteristics; Ashish Patel assisted by researching licensing and technical exchange agreements.

We offer our thanks to all these individuals for assisting us in assembling this significant amount of information. However, responsibility for any errors lies exclusively with the authors.

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FOREWORD

This text arose from a study originally undertaken for the Department of Energy to characterize the principal safety features of light water reactors of western design. It soon became apparent that these features could most effectively be presented by identifying them for each standard reactor type designed by a western reactor vendor. However, identification of these standard reactor types was not obvious because some vendors, notably Westinghouse for pressurized water reactors and ABB-Atom for boiling water reactors, did not give all their designs standard designations.

With the assistance of the relevant western reactor vendors and safety authorities, such an identification was made for all currently operating light water reactors during this study. For each operating plant we identify its family by assigning it to a standard type according to its principal features, and dates of order and initial commercial operation. Further, the history of licensing and technical exchange agreements among vendors has been described because such interchanges have contributed to significant parallelism in the design of the boiling light water reactors and in the design of pressurized light water reactors. The first third of the text is comprised of this information. The balance presents comparative technical descriptions of the principal safety features of each standard type of design.

This text should be of use to professional engineers interested in safety assessment of operating light water reactors, students interested in the principal safety features of LWRs, and others interested in tracing the design evolution of light water reactors. However, while ambitious in its scope, this text should not be viewed as presenting the levels of reactor safety of the various families of western reactor designs. Reactor safety encompasses a broad range of factors beyond design features that must be collectively considered to assess the level of safety, as described in Chapter 1.

The first, second and fourth authors of this text participated in this work in parallel with performing their thesis research while graduate students in the Department of Nuclear Engineering at the Massachusetts Institute of Technology (MIT). The third author, the Korea Electric Power Corporation Professor of Nuclear Engineering at MIT, directed this project.

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ABBREVIATIONS FOR ORGANIZATIONS

A-C	Allis Chalmers Co. (USA)
ABB-Atom	Asea Brown Boveri (Sweden/Switzerland)
ABB-CE NP	ABB-Combustion Engineering Nuclear Power (USA)
ABB-Reaktor	Asea Brown Boveri Reaktor (Germany)
Aceco	Ateliers de Construction, Electrique de Charleroi (Belgium)
Acecowen	Association des Ateliers de Charleroi et de Cockerill Ougree Province WNE/ Westinghouse (Belgium)
ACLF	Grouping of Acecowen/Creusot Loire (France)/Framatome/Westinghouse and Nuclear Europe (Belgium)
ACRS	Advisory Committee on Reactor Safeguards (USA)
AEE	Atomenergoexport (Russia)
AEG	AEG-Telefunken (Germany)
AEPSC	American Electric Power Service Co. (USA)
Agrupados	Empresarios Agrupados SA (Spain)
AMN	Ansaldo Meccanico Nucleare SpA (Italy)
ANA	Asociación Nuclear Ascó (Spain)
ANL	Argonne National Laboratory (USA)
ANVII	Asociación Nuclear Vandellós II (Spain)
APCO	Alabama Power Co. (USA)
APL	Arkansas Power and Light Co. (USA)
APS	Arizona Public Service Co. (USA)
ASME	American Society of Mechanical Engineers (USA)
Auxiesa	Auxini Ingeniera Espanola (Spain)
B&W	Babcock and Wilcox Service Co., Nuclear Power Division (USA)
Bayernwerk	Bayernwerk AG (Germany)
BBC	Brown Boveri et Cie (Switzerland)
BBR	Babcock Brown Boveri Reaktor (Germany)
Bechtel	Bechtel National, Inc. (USA)
BECO	Boston Edison Co. (USA)
BG&E	Baltimore Gas and Electric Co. (USA)
BKW	Bernische Kraftwerke AG (Switzerland)
Burns&Roe	Burns and Roe, Inc. (USA)
BV	Black and Veatch (USA)
CdT	Central de Trillo (Spain)
CE	Combustion Engineering, Inc., Nuclear Power Division (USA)
CEA	Commissariat à l'Energie Atomique (France)
CEIC	Cleveland Electric Illuminating Co. (USA)
CEN	Centre d'Etude de l'Energie Nucleaire (Belgium)
CFE	Cie d'Enterprises CFE (Belgium)
CFEM	Comisión Federal d'Electricidad (Mexico)
Chubu	Chubu Electric Power Co., Inc. (Japan)
Chugoku	The Chugoku Electric Power Co., Inc. (Japan)
CNA	Central Nuclear de Almaraz (Spain)
CNNC	China National Nuclear Corp. (China)

ComEd	Commonwealth Edison Co. (USA)
ConEd	Consolidated Edison Co. of New York, Inc. (USA)
CPC	Consumers Power Co. (USA)
CPL	Carolina Power and Light Co. (USA)
CYAP	Connecticut Yankee Atomic Power Co. (USA)
DE	Detroit Edison Co. (USA)
Duke	Duke Power Co. (USA)
Duquesne	Duquesne Light Co. (USA)
E&B	Emch & Berger (Switzerland)
Ebasco	Ebasco Services, Inc. (USA)
EdF	Electricité de France (France)
EI	Elettronucleare Italiana (Italy)
Electrabel	Compagnie Generale d'Enterprises Electriques et Industrielles (Belgium)
ENEL	Ente Nazionale per l'Energia Elettrica (Italy)
Ensa	Empresa Equipos Nucleares (Spain)
Enusa	Empresa Nacional del Uranio (Spain)
EPRI	Electric Power Research Institute (USA)
EPZ	NV Electriciteits-Productiemaatschappij Zuid (Netherlands)
Fenosa	Fuerzas Electricas del Noroeste, SA (Spain)
Fluor	Fluor Engineers and Contractors, Inc. (USA)
FPC	Florida Power Corp. (USA)
FPL	Florida Power & Light Co. (USA)
Fram	Framatome (France)
Framaceco	Framatome/ACEC/Cockerill (France)
Furnas	Furnas-Centrales Eléctricas SA (Brazil)
GA	General Atomics (USA)
GE	General Electric Company Nuclear Energy (USA)
Getsco	General Electric Technical Services Co. (USA)
G&H	Gibbs and Hill Co. (USA)
G&HE	Gibbs and Hill Española (Spain)
Gilbert	Gilbert Associates, Inc. (USA)
GKN ¹	Gemeinschaftskernkraftwerk Neckar GmbH (Germany)
GKW	Gemeinschaftskernkraftwerk Grohnde (Germany)
GNE	General Nuclear Engineering Co. (USA)
GNPJVC	Guangdong Nuclear Power Joint Venture Co. (China)
GPC	Georgia Power Co. (USA)
GPU Nuc	General Public Utilities Nuclear Corp. (USA)
GRS	Gesellschaft für Reaktorsicherheit (Germany)
GSU	Gulf State Utilities Co. (USA)
HE	Hidroelectrica Española (Spain)
Hitachi	Hitachi Ltd. (Japan)
HLP	Houston Light and Power Co. (USA)
Hochtief	Hochtief AG (Germany)
Hokkaido	The Hokkaido Electric Power Co., Inc. (Japan)
Hokuriku	Hokuriku Electric Power Co. (Japan)

¹ The names of the plant and operating organization are the same.

IAEA	International Atomic Energy Agency
IELP	Iowa Electric Light and Power Co. (USA)
Initec	Empresa Nacional de Ingeniera y Tecnologia (Spain)
INSAG	International Nuclear Safety Advisory Group
IPC	Illinois Power Co. (USA)
IVO	Imatran Voima Oy (Finland)
JAPCO	The Japan Atomic Power Co. (Japan)
KAERI	Korean Atomic Energy Research Institute (Korea)
Kajima	Kajima Construction Co. (Japan)
Kansai	Kansai Electric Power Co., Inc. (Japan)
KBR	Kernkraftwerk Brokdorf (Germany)
KEPCO	Korea Electric Power Corporation (Korea)
KGB	Kernkraftwerk Gundremmingen Verwaltungsgesellschaft (Germany)
KKB ¹	Kernkraftwerk Brunsbüttel (Germany)
KKG-D	Kernkraftwerk Gösgen-Daniken AG (Switzerland)
KKI ¹	Kernkraftwerk Isar (Germany)
KKK ¹	Kernkraftwerk Krümmel (Germany)
KKL	Kernkraftwerk Leibstadt AG (Switzerland)
KKP ¹	Kernkraftwerk Philippsburg GmbH (Germany)
KKS ¹	Kernkraftwerk Stade (Germany)
KKU ¹	Kernkraftwerk Unterweser (Germany)
KLE	Kernkraftwerk Lippe-Ems (Germany)
KOPEC	Korea Power Engineering Co. (Korea)
KWG	Kernkraftwerk Grohnde (Switzerland)
KWO ¹	Kernkraftwerk Obrigheim GmbH (Germany)
KWU	Kraftwerk Union AG (Germany)
Kyushu	Kyushu Electric Power Co., Inc. (Japan)
LPL	Louisiana Power and Light Co. (USA)
Mapi	Mitsubishi Atomic Power Industries (Japan)
MHI	Mitsubishi Heavy Industries (Japan)
MITI	Ministry of International Trade and Industry (Japan)
MYAPC	Maine Yankee Atomic Power Co. (USA)
NEK	Nuclearna Elektrarna Krsko (Slovenia)
NEU	Northeast Utilities Service Co. (USA)
NMPC	Niagara Mohawk Power Corp. (USA)
NNC	National Nuclear Corp. (UK)
NOK	Nordöstscheizerische Kraftwerk AG (Switzerland)
NPC	Nuclear Power Corp. (India)
NPI	Nuclear Power International (France and Germany)
NPPD	Nebraska Public Power District (USA)
NSP	Northern States Power Co. (USA)
Nuclen	Nuclebras Engenharia SA (Brazil)
Nuclenor	Centrales Nucleares del Norte (Spain)
NVGKN	NV Gemeenschappelijke Kernenergiecentrale Nederland (Netherlands)
NYP&A	New York Power Authority (USA)
OKG	Oskarshamnvernets Kraftgrupp AB (Sweden)

¹ The names of the plant and operating organization are the same.

OPPD	Omaha Public Power District (USA)
OPS	Offshore Power Systems Co. (USA)
PECO	Philadelphia Electric Co. (USA)
PG&E	Pacific Gas & Electric Co. (USA)
PP&L	Pennsylvania Power and Light Co. (USA)
Preag	Preussen Elektra AG (Germany)
Promon	Promon Engenharia (Brazil)
PRWRA	Puerto Rico Water Resources Authority (Puerto Rico)
PS&E	Pioneer Service and Engineering Co. (USA)
PSEG	Public Service Electricity and Gas Co. (USA)
RGE	Rochester Gas and Electric Co. (USA)
Ruhrstahl	Ruhrstahl Apparatebau (Germany)
RWE	Rheinisch-Westfälisches Elektrizitätswerk Energie AG (Germany)
S&P	Sverdrup & Parceo (USA)
Sarg&Lun	Sargent and Lundy Engineers (USA)
SCE&G	South Carolina Electric and Gas Co. (USA)
SEBIM	Société d'Exploitation des Brevets pour l'Industrie et la Marine (France)
SENA	Société d'Energie Nucléaire Franco-Belge des Ardennes (France, Belgium)
SERI	System Energy Resources, Inc. (USA)
Shikoku	Shikoku Electric Power Co. (Japan)
Siemens	Siemens AG (Germany)
SoCal Ed	Southern California Edison Co. (USA)
SPO-ND	Siemens Power Operations, Nuclear Division (Germany)
SS	Southern Services Co. (USA)
SSPB	Swedish State Power Board (Sweden)
Stone&Web.	Stone and Webster Engineering Corp. (USA)
Sydskraft	Sydsvenska Kraftaktibolaget (Sweden)
TEPCO	Tokyo Electric Power Co., Inc. (Japan)
Toshiba	Toshiba Corp. (Japan)
Tohoku	Tohoku Electric Power Co., Inc. (Japan)
TolEd	Toledo Edison Co. (USA)
TPC	Taiwan Power Co. (Taiwan)
TUE	Texas Utilities Electric Co. (USA)
TÜV	Technischer Überwachungs Verein (Germany)
TVA	Tennessee Valley Authority (USA)
TVO	Teollisuuden Voima Oy (Finland)
UE	Union Electric Co. (USA)
UEC	United Engineers and Constructors Co. (USA)
UN	United Nations
USAEC	United States Atomic Energy Commission (USA)
USDOE	United States Department of Energy (USA)
USNRC	United States Nuclear Regulatory Commission (USA)
VBB	Vattenbyggnadsbyran (Sweden)
V-O	Voima Oy (Finland)
VP	Virginia Power (USA)
VYNPC	Vermont Yankee Nuclear Power Corp. (USA)
WCNOC	Wolf Creek Nuclear Operating Corp. (USA)

WEP	Wisconsin Electric Power Co. (USA)
West.	Westinghouse (USA)
WPS	Wisconsin Public Service Corp. (USA)
WPPSS	Washington Public Power Supply System (USA)

ACRONYMS

1-loop	Standard Westinghouse Pressurized Water Reactor Type
2-loop	Standard Westinghouse, B&W and Siemens Pressurized Water Reactor Types
3-loop, 4-loop	Standard Westinghouse and Siemens Pressurized Water Reactor Types
ABWR	Advanced Boiling Water Reactor
ac	Alternating Current
ADS	Automatic Depressurization System
A-E	Architect-engineer
AFA	Advanced Fuel Assembly
AFS	Auxiliary Feedwater System
AGR	Advanced Gas-cooled Reactor
ALWR	Advanced Light Water Reactor
AP600	Advanced Plant 600 MWe Reactor
APWR	Advanced Pressurized Water Reactor
ARSAP	Advanced Reactor Severe Accident Program
ATWS	Anticipated Transient Without Scram
BOC	Beginning of Cycle
BONUS	Boiling Nuclear Superheat Reactor, Puerto Rico (USA)
BORAX	Boiling Water Reactor Experiment (USA)
BWR	Boiling Water Reactor
BWR/1, 2, 3, 4, 5, 6	Standard General Electric Boiling Water Reactor Types
BWR/69, 72	Standard Siemens Boiling Water Reactor Types
BWR/G1, G2, G3, 75	Standard ABB Boiling Water Reactor Types
CHFR	Critical Heat Flux Ratio
CHW	Condenser Hotwell
CP0, 1, 2	Standard Framatome Pressurized Water Reactor Types
CPR	Critical Power Ratio
CRBR	Controlled-recirculation Boiling Water Reactor; also called Pathfinder (USA)
CRDM	Control Rod Drive Mechanism
CSS	Containment Spray System
CST	Condensate Storage Tank
CVCS	Chemical and Volume Control System
CWS	Cooling Water System
DBA	Design Basis Accident
dc	Direct Current
DCH	Direct Containment Heating
DF	Decontamination Factor
DNB	Departure from Nucleate Boiling
DNBR	Departure from Nucleate Boiling Ratio
EBWR	Experimental Boiling Water Reactor (USA)
ECCS	Emergency Core Cooling System
EFWS	Emergency Feedwater System
EPR	European Pressurized Water Reactor

ESF	Engineered Safeguard Feature or Engineered Safety Feature
EVESR	Esada Vallecitos Experimental Superheat Reactor (USA)
FCI	Feedwater Coolant Injection
FILTRA	Trademark for ABB-Atom Filtered Venting System
FSAR	Final Safety Analysis Report
GDC	General Design Criteria
GKN ¹	Gemeinschafts Kernkraftwerk Neckar GmbH (Germany)
HHSI	High-Head Safety Injection
HPCI	High-Pressure Coolant Injection
HPCS	High-Pressure Core Spray
IASCC	Irradiation-assisted Stress Corrosion Cracking
IC	Isolation Condenser
ICS	Integrated Control System
ICSLOCA	Interfacing Coolant System Loss of Coolant Accident
IFM	Intermediate Flow Mixing Vane Grid
IGSCC	Intergranular Stress Corrosion Cracking
IRHRS	Independent Residual Heat Removal System
JPDR	Japan Power Demonstration Reactor (Japan)
KBR	Kernkraftwerk Brokdorf (Germany)
KKB ¹	Kernkraftwerk Brunsbüttel (Germany)
KKE	Kernkraftwerk Emsland (Germany)
KKG	Kernkraftwerk Grafenrheinfeld (Germany)
KKH	Kernkraftwerk Hamm (Germany)
KKI ¹	Kernkraftwerk Isar (Germany)
KKK ¹	Kernkraftwerk Krümmel (Germany)
KKN	Kernkraftwerk Niederaichbach (Germany)
KKP ¹	Kernkraftwerk Philippsburg GmbH (Germany)
KKS ¹	Kernkraftwerk Stade (Germany)
KKU	Kernkraftwerk Unterweser (Germany)
Konvoi	Standard Siemens Pressurized Water Reactor Type
KRB	Kernkraftwerk Gundremmingen (Germany)
KWB	Kernkraftwerk Biblis (Germany)
KWG	Kernkraftwerk Grohnde (Switzerland)
KWL	Kernkraftwerk Lingen (Germany)
KWO ¹	Kernkraftwerk Obrigheim GmbH (Germany)
KWS	Kernkraftwerk Wyhl, Süd (Germany)
KWW	Kernkraftwerk Würgassen (Germany)
LHSI	Low Head Safety Injection
LOCA	Loss of Coolant Accident
LPCI	Low-Pressure Coolant Injection
LPCS	Low-Pressure Core Spray
LPSI	Low-Pressure Safety Injection
LSSS	Limiting Safety System Settings
LWR	Light Water Reactor
MCPR	Minimum Critical Power Ratio
MDNBR	Minimum Departure from Nucleate Boiling Ratio

¹ The names of the plant and operating organization are the same.

MK-I, -II, -III	Mark-I, -II, -III BWR Containments
MSIV	Main Steam Isolation Valve
MVSS	Multi Venturi Scrubber System
N4	Standard Framatome Pressurized Water Reactor Type
NA	Not Applicable
NE	Nonexistent
NPP	Nuclear Power Plant
NS	Not Selected
NSSS	Nuclear Steam Supply System
Nuplex 80+	Integrated Control System of the S80+ Reactors
OFA	Optimized Fuel Assembly
OTSG	Once-through Steam Generator
P4, P'4	Standard Framatome Pressurized Water Reactor Types
PCI	Pellet-cladding Interaction
PCS	Plant Control System
PHWR	Pressurized Heavy Water Reactor
PIUS	Process Inherent Ultimate Safety Reactor
PORV	Pilot- or Power-operated Relief Valve
Pre-Konvoi	Several similar but not standard Siemens Pressurized Water Reactors that preceded the Konvoi reactor type
Pre-System 80 (Pre-S80)	Standard Combustion Engineering Pressurized Water Reactor Type
Pres Supp	Pressure Suppression Type of Containment System
PTS	Pressurized Thermal Shock
PWR	Pressurized Water Reactor
RCIC	Reactor Core Isolation Cooling
RCFCS	Reactor Containment Fan Cooler System
RCS	Reactor Coolant System
RHRS	Residual Heat Removal System
RPS	Reactor Protection System
RWST	Refueling Water Storage Tank
SBWR	Simplified Boiling Water Reactor
SEHR	Special Emergency Heat Removal
SIS	Safety Injection System
SL-1	Stationary Low Power Reactor 1 (USA)
SLCS	Standby Liquid Control System
SNUPPS	Standardized Nuclear Utility Power Plant System
SPC	Suppression Pool Cooling
SP90	Safe Pressurized 90
SVEA	“Water-cross” Fuel Assembly (ABB-Atom)
System 80 (S80)	Standard ABB-Combustion Engineering Pressurized Water Reactor Type
System 80+ (S80+)	Standard ABB-Combustion Engineering Pressurized Water Reactor Type
System 80K (S80K)	Upgraded ABB-Combustion Engineering System 80 standardized in cooperation with the Korean Electric Power Corporation (KEPCO)
TBD	To Be Determined
TMI	Three Mile Island (USA)
USSR	Union of Soviet Socialist Republics

UTSG	U-tube Steam Generator
V5	Vantage 5 Westinghouse Fuel Assembly
VAK	Versuchsatomkraftwerk Kahl (Germany)
VBWR	Vallecitos Boiling Water Reactor (USA)
VVER	Vodo-Vodjanoy Energeticeskiy Reaktor (Russian pressurized water reactor)
WNP	Washington Nuclear Project, BWR/5 (USA)

SYMBOLS USED IN DRAWINGS

MECHANICAL:

	Open Valve			Pumps
	Closed Valve			High-pressure Tank or Vessel
	Motor-operated Valve			Tanks
	Check Valves			Sump
	Relief Valve			Steam Generators
	Three Way Valve			Steam Drum
	Differential Pressure Valve			Turbine
	Magnetically-operated Pilot Valve			Condensers
	Spring-operated Pilot Valve			Heat Exchangers
	Vacuum Breaker			
	Rupture Disk			
	Orifices			
	Spray			

ELECTRICAL:

	Batteries		Generator
	Battery Charger		Motor
	Static Inerter		Diesel
	Motor-operated Disconnect		Transformers
	Circuit Breakers, opened/closed		
	Switches, opened/closed		



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Chapter 1

INTRODUCTION

1.1 SCOPE

Reactor safety encompasses a broad range of factors including design, construction, operation, and training. Effecting and integrating all is the safety culture that governs the actions and interactions of all individuals and organizations operating the plants.

The means employed to provide reactor safety have continually evolved from the designs of the first reactors that tested the feasibility of nuclear power generation to the advanced designs that are currently under development. More specifically, the premises underlying reactor safety have evolved as vulnerabilities of previous designs were uncovered. For example, western designers recognized early the necessity of being able to prevent potential releases of radioactivity to the environment by using a containment building. The need for increased reliability of normal operating systems and components was also recognized early as a significant contributor to improved reactor safety.

Early in the development of light water reactors (LWRs), assurance of adequate control of reactivity was among the most important considerations. The safe behavior of light water reactors with low-enrichment uranium oxide fuel during a very large, rapid power increase was demonstrated by experiments in which the transient was terminated by the negative Doppler effect on reactivity. To relax siting requirements regarding the distance between the reactor and the population of neighboring communities, engineering design approaches evolved in the form of very low leakage containments, containment spray systems and containment atmosphere cleanup systems. These systems were called engineered safeguards. These safeguards were subsequently supplemented by other systems (e.g., the emergency core cooling system) and the collective grouping became also known as engineered safety features (ESFs). Later, as reactor power ratings were substantially increased, attention was focused on the consequences of the large break loss of coolant accident (LOCA). This prompted requirements for enhanced capabilities in the design of emergency core cooling systems (ECCSs) and for further improvements in the preservation of the primary system pressure boundary.

In addition to the continual safety enhancement achieved through improvement of normal operating systems and components as well as expansion of ESFs, reactor safety has further evolved by the introduction of special features that enhance the plant's ability to cope with abnormal occurrences. These special features are a diverse collection of systems and components that range from special power monitoring systems to special features of the containment. Okrent, 1981, provides a detailed history of the evolution of light water reactor safety that introduces these as well as other salient events through examination of the deliberations of the USA's Advisory Committee on Reactor Safeguards (ACRS).

Some of the main events that led to significant changes in reactor safety design were:

- 1957 – The Windscale accident emphasized the need for adequate containment because there are not enough remote sites suitable for construction of nuclear reactors. [Thompson and Beckerley, 1964] (Containments had already been used for several reactors at that time.)
- 1961 – The Stationary Low Power Reactor 1 (SL-1) accident led to design requirements on reactivity control—shutdown margin, control rod worth, and stability. [Thompson and Beckerley, 1964]
- 1967 – The Ergen Task Force reconsidered the worst-case accident scenario and its consequences, given the increasing size and power ratings of USA LWRs. [Emergency Core Cooling, 1967] This triggered modernization of the design of ECCSs.
- 1975 – The Browns Ferry fire emphasized the need for separation of redundant electrical networks. [Rust and Weaver, 1976]

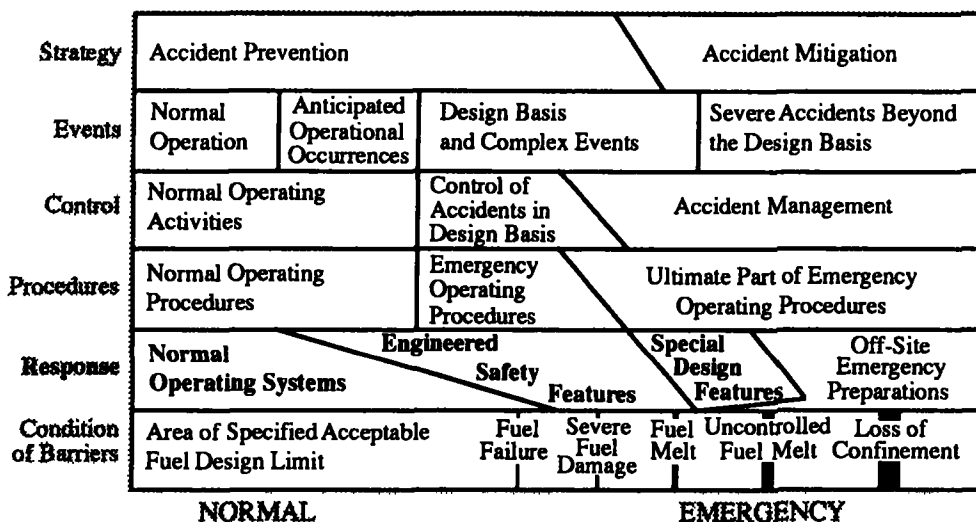


Figure 1-1. Overview of Defense-in-depth Concept (taken from INSAG, 1988).

- 1975 – The Reactor Safety Study showed that core damage and containment failure could follow station blackouts and small break LOCAs. [WASH-1400, 1975] It introduced the probabilistic risk assessment methodology to the nuclear industry, which has since gained recognition as a powerful tool for assessing safety and safety improvements.
- 1979 – The Three Mile Island accident directed attention toward severe-accident considerations and improvement of the man-machine interface. [Kemeny, 1979]
- 1986 – The Chernobyl accident stimulated concern about beyond-design-basis accidents and led many European countries to consider using designated containment venting systems. [NUREG-1250, 1987]

The International Atomic Energy Agency (IAEA), through its International Nuclear Safety Advisory Group (INSAG), has developed a set of principles for the safe management and operation of reactor systems. [INSAG, 1988] Among them, the defense-in-depth concept was singled out by INSAG as fundamental to the safety technology of nuclear power. This concept promotes the enhancement of safety by the erection of multiple and diverse barriers to the release of fission products to the environment. It includes design features and measures to protect the barriers themselves and to protect the public and the environment in case these barriers are not fully effective.

Figure 1-1 was developed by INSAG to provide an overview of the defense-in-depth approach. Highlighted there under “Response” are features that contribute to the defense-in-depth approach to reactor safety:

- Normal operating systems,
- Engineered safety features, and
- Special design features.

It is these systems and features that are the subject of this volume. The presentation of design features in this volume is organized to illustrate the defense-in-depth concept and their contribution to either accident prevention or mitigation.

The coverage has been confined to operating LWRs of western design. Substantial numbers of heavy water reactors of Canadian and Indian design as well as gas cooled reactors of British design are operating; the focus here is on LWRs because they are more numerous by far. Pressurized light water reactors have also been extensively developed in the former USSR and deployed both there and in countries of eastern Europe; however, those pressurized water reactors are of substantially different design than their western counterparts, so their inclusion would have

required major additions to the text. Again to make the scope manageable, only operating LWRs are covered although considerable work worldwide is underway on advanced reactor designs. Such new directions are identified as appropriate to maintain the continuity of the discussion of design evolution, but the safety characteristics of these advanced designs are not treated comprehensively.

Finally, it is beyond the scope of this work to quantify the safety contributions of the features covered; quantification would require definition of all system details and quantitative assessments of risk. Furthermore, the technical basis and utilization of certain safety concepts fundamental to the configuration of current LWR designs (for example, the exclusion of reactor vessel failure and occurrence of leak-before-break for specific categories of piping) are also not presented.

1.2 EVOLUTION OF DESIGNS

The LWR is the reactor concept that has been most widely developed and utilized throughout the world. The two manifestations of this concept, the boiling water reactor (BWR) and the pressurized water reactor (PWR), were originally designed by United States companies and constructed for USA utilities. Subsequent extrapolation of these concepts involved complex commercial agreements that led to both direct international sales of USA reactor designs and development of distinct designs by other international companies.

For the BWR, in Germany AEG-Kraftwerk Union (KWU), currently Siemens, evolved their own reactor designs subsequent to an initial licensing agreement with General Electric (GE). In Sweden, Asea-Atom, currently ABB-Atom, independently developed several successive BWR designs. Hitachi and Toshiba of Japan began their BWR program by building GE-licensed reactors. Subsequent independent Japanese development of the BWR concept focused primarily on improving ergonomics and on containment design. Their most important recent contributions are as collaborators with GE in the design of the advanced boiling water reactor (ABWR).

For the PWR, following initial arrangements with Westinghouse, France's Framatome and to a larger extent Germany's Siemens developed original designs. In the USA, Babcock & Wilcox (B&W) and Combustion Engineering (CE), currently ABB-CE NP, have also developed original PWR designs. Japan's PWR designer, Mitsubishi Heavy Industries (MHI), began by building Westinghouse-licensed PWRs. Through Japan's independent LWR Improvement and Standardization Program, Mitsubishi contributed to extending the use of microcomputer technology in their currently operating PWRs and to improving the man-machine interface. Mitsubishi's recent contributions to PWR evolution are evident in the advanced designs on which they are collaborating with Westinghouse.

1.3 SAFETY THROUGH DESIGN

The enhancement of safety through design is typically accomplished by three means:

- Selection of a neutronic design that will inherently decrease core reactivity under abnormal conditions that could lead to overheating—this characteristic is termed inherent negative reactivity feedback,
- Adoption of simple, rugged design approaches for key components and systems, and
- Provision for diverse and redundant means for accomplishing critical safety functions.

A prime example of the first means is the negative Doppler reactivity coefficient of the fuel in all light-water-moderated power reactors. The negative void coefficients in BWRs and PWRs¹ are other examples. The fuel Doppler coefficient arises because the variation of capture probability in U-238 with neutron energy exhibits very high but narrow peaks. Raising fuel temperature, and

¹ PWRs experience only subcooled boiling during normal operation; hence void reactivity feedback is unimportant during normal operations.

hence the velocities of the U-238 nuclei, in effect broadens this range and hence increases the neutron capture rate. The void coefficient arises from the decrease in density of the coolant moderator as its temperature increases. This increases the leakage out of the core and into control rods, and also shifts the neutron spectrum to higher energies, hence decreasing the fission rate in U-235. Decreasing the moderator density also increases resonance neutron absorption in U-238.

The second means of enhancing safety through design has been the adoption of simple, rugged design approaches for a variety of key components and systems. Examples include the designs of control rod drive mechanisms, fuel arrays, jet pumps, internal pumps, and natural circulation capability.

Key features of the nuclear steam supply system (NSSS) that fall in these first two categories are presented in Chapter 4.

Achievement of safety, however, is safeguarded further by enhancing the means available to perform a critical safety function no matter how well the individual operating mechanism has been designed; this enhancement is accomplished by either: (a) redundancy of the number of mechanisms of a given design that can perform the function, or (b) diversification of the operating principles upon which the mechanisms perform their safety functions. A prime example of diversity is the use of two different means of shutdown; adoption of design diversity ensures performance of critical safety functions by minimizing the susceptibility to common-mode failure.

These strategies of redundancy and diversity are not mutually exclusive, but are in fact most often utilized in complementary fashion. Examples of the simultaneous use of redundancy and diversity abound in implementation of engineered safeguard systems designed to prevent and mitigate the consequences of assumed accident initiators. Detailed presentations of redundant and diverse systems incorporated in various standard reactor types appear in Chapters 5 to 7. It should be recognized that the redundancies are in the number of trains of systems including their power supplies since each train depends on the availability of electric power for its operation.

1.4 REDUNDANCY CLASSIFICATION

Explanation of the classification scheme for specifying the degree of redundancy for safety-related systems will be useful for understanding the later detailed presentations of plant designs, particularly with regard to electrical and fluid systems.

This general classification scheme for redundancy is designated by the notation $N + \#$. The nominal achievement of a safety function can be designed to be accomplished by a single or by multiple delivery systems. For example, residual heat removal can be accomplished by either a single 100%-capacity train or by two 50%-capacity trains. In these cases the value of N is 1 or 2, respectively.

The subsequent digit, “#”, describes the number of trains beyond N that are provided to achieve a desired degree of redundancy. In general, this redundancy can be provided to accommodate:

- One train out of service for maintenance, or
- One train failing to perform its function either because of failure to start or because of a component fault.

If two additional trains were provided, one explicitly for each of the above eventualities, then the design would be characterized as an $N + 2$ design. Further, if N were 2, i.e., two 50% trains, then the total number of identical but separate trains would be 4. Such an $N + 2$ plant can be operated with a train out of service for maintenance or repair. This can potentially contribute to the reduction of plant downtime. If the plant design is based on the $N + 1$ redundancy scheme, then only a single additional train is provided. In this case, the plant cannot be operated while a train is out for maintenance because the additional failure of another train would render the overall system unable to perform its intended function.

Actual implementation of a design scheme can involve selections that cannot be described fully by this simple $N + \#$ scheme. For example, because systems use active (e.g., pumps) as well as

passive (e.g., heat exchangers) components that typically have histories of different reliabilities, separate portions of a system can be designed with different degrees of redundancy. Further, certain components, say a combustion turbine for pumping, can be provided in the plant for possible use with a number of otherwise independent trains; hence, such a component is characterized as a “swing” component. Finally, arrangement of the electrical distribution system can sometimes allow for cross-tying, which changes the degree of independence of a given number of trains.

It would have been desirable to contrast standard plant types by the “N + #” redundancy approach, particularly in this cross-national comparison effort. That is not possible for three reasons:

1. The evolution of reactor plant design worldwide has led to continuing change in the degree of design redundancy selected; this evolutionary trend has been to increase redundancy unevenly with more redundancy provided for main frontline safety systems (i.e., safety injection, emergency feedwater and on-site electrical) than for auxiliary systems of safety significance.
2. Within a given design organization, advantage has been taken of the nature of different components, leading to varying degrees of cross-tying.
3. Due to different safety philosophies, some countries have addressed the common-mode failure issue by different combinations of isolated, bunkered trains and the addition of redundant trains.

Even among plants of the same standard type, diversity and redundancy considerations have been implemented differently. For these reasons, the standard types cannot each be characterized by the simple “N + #” designation. However, the general characterization can be made that LWR plants range from N + 1 to N + 2 in the following approximate order of country of origin—the USA and France, to Sweden, and to Germany.

1.5 ORGANIZATION OF THE TEXT

The first part of the text covers the classification of operating BWRs and PWRs into standard design types. The process of classification begins with the identification of the western NSSS vendors that made original contributions to LWR design, i.e., the original designers. Then the standard types of reactors produced by each original designer are identified and classified by their predominant NSSS features.

Chapter 2 lists all operating (as of June 1994) LWRs of western design; both BWRs and PWRs are listed according to the defined standard types. Chapter 2 also includes diagrams that show all the operating BWRs and PWRs of each country. These diagrams, which will be referred to as “family trees,” show the original designer and standard type of each unit on the horizontal axis, and the year of order on the vertical axis.

Chapter 3 presents the chronological development of the standard LWR types produced by each original designer. The principal features of each standard type are presented. Also shown are the time frames of licensing agreements and technical exchanges among the original designers.

The second part of the text (Chapters 4 to 9) focuses on safety-related features of currently operating LWRs. The design evolutions of the systems and components are presented and, wherever pertinent, the motives and implications of the advances are detailed. Attention is focused on the safety-related differences among various standard types of western BWRs and among various standard types of western PWRs. These differences are partially due to the specific design requirements within a particular country, which influence the design approach of the original designer. Furthermore, the response of each designer to operating experience has contributed to differences among successive standard types and, sometimes, among successive

plants of the same type. This trend of design improvement based on operating experience continues in the development of advanced LWRs today.

Descriptions of key safety-related design features of the NSSS are grouped together in Chapter 4; the evolution of each of these features and the consequent differences among successive LWR types are discussed. Electrical systems, the plant protection system, emergency coolant and post-accident heat removal systems, and containment are covered in individual, subsequent chapters. The electrical network layout and the reactor protection system logic of each LWR standard type are covered in Chapters 5 and 6, respectively. Emergency coolant inventory makeup and post-accident heat removal systems are the subject of Chapter 7; special emphasis is placed on the redundancy and diversity of the safety injection and recirculation systems adopted for each standard type. The various containment designs and associated systems are described in Chapter 8. Chapter 9 identifies design features significant to the consideration of severe reactor accidents. Chapter 10 summarizes the most important differences among the existing standard BWR and PWR designs. It further identifies the current safety-related trends in LWR design through examples of features in advanced plants currently under construction or development.

It should be noted that there are safety-related features (e.g., neutron monitoring systems, radiation monitoring system sensors) that are not covered because they are either notably similar among the various standard types of LWRs or they have not undergone significant design evolution. Dimensions and numerical characteristics of many important safety features are included in the chapters where these features are presented. However, these dimensions together with others for basic reactor design features have not been compiled in summary form for this text.¹

1.6 SUMMARY

This chapter defines the scope of the text and outlines the organization of its chapters. The scope is confined to systems and design features that contribute to the defense-in-depth approach to reactor safety. Attention is confined to operating light water reactors of western design. These are the boiling water reactor and pressurized water reactor, for which standard design types have been produced by USA, German and Swedish vendors, and by USA, French and German vendors, respectively.

Safety has been achieved by incorporating inherent negative reactivity feedback, by using simple, rugged design approaches, and by providing for diverse and redundant means for accomplishing critical safety functions. The strategies of redundancy and diversity are not mutually exclusive, but have often been used in complementary fashion.

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Chapter 2

REACTOR FAMILY TREES

The currently operating LWRs, both BWRs and PWRs, can be classified into standard types based on their principal technical features. Although such an identification has not been made by each design firm for its reactors, we use the existing designations for reactor types that have been so identified; in the remaining cases, we have made new classifications according to predominant nuclear steam supply system (NSSS) characteristics with the assistance of the relevant¹ reactor vendors and safety authorities. This chapter presents the most important safety features for each of the defined standard types of BWRs and PWRs.

The variation of safety features within the family of plants of each standard type of BWR or PWR depends on the degree of standardization adopted by the particular designer or imposed by the utility customer. For example, there are practically no differences among PWRs of French design that belong to the same type (or series as designated by Electricité de France (EdF)). At the other extreme are many USA LWRs (both BWRs and PWRs) for which two plants of the same type can exhibit large differences between their safety-related features; these differences result from two principal factors:

1. The architect-engineer (A-E) or the utility or both expressed specific preferences at the time of the initial NSSS design.
2. Technological evolution and regulatory changes during construction led to the implementation of features that, for various reasons, were not backfitted uniformly into all the previous plants of that particular type.

Nevertheless, the benefits of standardization were being recognized in the USA by the early 1970s resulting in three PWR plant groups each of identical design. These included eight of the last twenty-six plants (six BWRs, twenty PWRs) ordered and completed in the USA.

2.1 STANDARD TYPES OF BWRs

Three countries have been active in developing BWRs: the USA, Germany, and Sweden. The major reactor designers are General Electric (GE), Siemens,² and ABB-Atom,³ respectively. Each designer has produced successive types of reactors that have sequentially incorporated significant technological advances. The history of these designers and their original contributions to the development of the BWR concept are the scope of this section. Toshiba and Hitachi of Japan have contributed in the improvement of the containments and the man-machine interfaces; Japanese contributions to BWR containment design will be mentioned in the relevant sections.

BWRs of western design can be classified as belonging to one of the standard types listed in Table 2.1. The principal differences among these BWR types pertain to the design of reactor coolant systems, emergency systems, and auxiliary systems.

Among the most significant differences in design features within each of the standard BWR types is the containment design, which has itself progressed through a series of standard designs. Consequently, there are different combinations of standard reactor types and standard containment

¹ The Swedish designs prior to the BWR/75, the German PWRs prior to Konvoi and the Westinghouse designs prior to SNUPPS.

² Kraftwerk Union AG was established in 1969 with 50/50 participation of AEG-Telefunken and Siemens. In the 1980s AEG sold its share of KWU to Siemens, which led to the consolidation of KWU into Siemens. Thus, KWU is an Operating Group of Siemens Corporation that designs both BWRs and PWRs.

³ Asea-Atom was established in 1969 with 50/50 ownership by the Swedish government and the Asea group. In 1981 Asea became the sole owner of Asea-Atom. In 1988 ABB-Atom was formed by merger of Asea with the German/Swiss company Brown Boveri. (See Figure 3-3 for the evolution of ABB's nuclear divisions, including that of the BWR designer, ABB-Atom.)

Table 2.1 Standard Types of Operating BWRs of Western Design

Country	Original designer	Type
USA	General Electric	BWR/1 ^a BWR/2 BWR/3 BWR/4 BWR/5 BWR/6
Germany	Siemens	BWR/69 BWR/72
Sweden	ABB-Atom	BWR/G1 ^b BWR/G2 BWR/G3 BWR/75

^a Designation given to GE designs preceding the standardization of the **BWR/2** concept.

^b **BWR/G1, 2, 3**,—i.e., Generation 1, 2, 3—are designations given to BWRs prior to the **BWR/75** standard type, which have not been formally named by ABB-Atom.

NOTE: All standard types will appear in bold typeface throughout the volume.

designs. Section 2.3 contains the family trees of both BWRs and PWRs for countries with LWR reactors of western design; the standard BWR containment design for each plant is also included in Section 2.3.

Table 2.2 lists all BWRs operating as of June 1994. This table identifies the A-E and the NSSS vendor as well as the utility operating the plant. Appendix A lists BWRs that have been shut down, are under construction, or have been cancelled, deferred or suspended.

Table 2.2 Boiling Water Reactors (operating as of June 1994)^a

Plant name	Type	Architect-engineer	NSSS vendor ^b	Operator
FINLAND (2 units)				
Olkiluoto 1	BWR/G3	ABB-Atom (Asea-Atom) ^c	ABB-Atom (Asea-Atom)	TVO
Olkiluoto 2	BWR/G3	ABB-Atom (Asea-Atom)	ABB-Atom (Asea-Atom)	TVO
GERMANY (7 units)				
Brunsbüttel (KKB)	BWR/69	Siemens (KWU)	Siemens (KWU)	KKB
Gundremmingen B (KRB-B)	BWR/72	Siemens (KWU)	Siemens (KWU)	KGB
Gundremmingen C (KRB-C)	BWR/72	Siemens (KWU)	Siemens (KWU)	KGB
Isar 1 (KKI-1)	BWR/69	Siemens (KWU)	Siemens (KWU)	KKI
Krümmel (KKK)	BWR/69	Siemens (KWU)	Siemens (KWU)	KKK
Philippsburg 1 (KKP-1)	BWR/69	Siemens (KWU)	Siemens (KWU)	KKP
Würgassen (KWW)	BWR/69	Siemens (AEG)	Siemens (AEG)	Preag
INDIA (2 units)				
Tarapur 1	BWR/1	Bechtel	GE	NPC
Tarapur 2	BWR/1	Bechtel	GE	NPC
JAPAN (24 units)				
Fukushima Daiichi 1	BWR/3	Ebasco	GE	TEPCO
Fukushima Daiichi 2	BWR/4	Ebasco	GE/Toshiba	TEPCO
Fukushima Daiichi 3	BWR/4	Toshiba	Toshiba	TEPCO
Fukushima Daiichi 4	BWR/4	Hitachi	Hitachi	TEPCO
Fukushima Daiichi 5	BWR/4	Toshiba	Toshiba	TEPCO
Fukushima Daiichi 6	BWR/5	Ebasco	GE/Toshiba	TEPCO

Table 2.2 Boiling Water Reactors (operating as of June 1994)^a (continued)

Plant name	Type	Architect-engineer	NSSS vendor ^b	Operator
JAPAN (continued)				
Fukushima Daini 1	BWR/5	Toshiba	Toshiba	TEPCO
Fukushima Daini 2	BWR/5	Hitachi	Hitachi	TEPCO
Fukushima Daini 3	BWR/5	Toshiba	Toshiba	TEPCO
Fukushima Daini 4	BWR/5	Hitachi	Hitachi	TEPCO
Hamaoka 1	BWR/4	Toshiba	Toshiba	Chubu
Hamaoka 2	BWR/4	Toshiba/Hitachi	Toshiba/Hitachi	Chubu
Hamaoka 3	BWR/5	Toshiba/Hitachi	Toshiba/Hitachi	Chubu
Hamaoka 4 ^d	BWR/5	Toshiba/Hitachi	Toshiba	Chubu
Kashiwazaki-Kariwa 1	BWR/5	Toshiba	Toshiba	TEPCO
Kashiwazaki-Kariwa 2	BWR/5	Toshiba	Toshiba	TEPCO
Kashiwazaki-Kariwa 3 ^e	BWR/5	Toshiba	Toshiba	TEPCO
Kashiwazaki-Kariwa 5	BWR/5	Hitachi	Hitachi	TEPCO
Onagawa 1	BWR/4	Kajima	Toshiba	Tohoku
Shika 1 ^f	BWR/4	Hitachi	Hitachi	Hokuriku
Shimane 1	BWR/3	Hitachi	Hitachi	Chugoku
Shimane 2	BWR/5	Hitachi	Hitachi	Chugoku
Tokai 2	BWR/5	Ebasco	GE/Hitachi	JAPCO
Tsuruga 1	BWR/2	Ebasco	GE/Hitachi	JAPCO
MEXICO (1 unit)				
Laguna Verde 1	BWR/5	Ebasco/CFE	GE	CFEM
NETHERLANDS (1 unit)				
Dodewaard	BWR/1	NVGKN	GE/GKN	NVGKN
SPAIN (2 units)				
Cofrentes	BWR/6	Agrupados/G&H	GE	HE
Santa Maria de Garoña	BWR/3	Ebasco	GE	Nuclenor
SWEDEN (9 units)				
Barsebäck 1	BWR/G2	ABB-Atom (Asea-Atom) ^{c,g}	ABB-Atom (Asea-Atom)	Sydkraft
Barsebäck 2	BWR/G2	ABB-Atom (Asea-Atom)	ABB-Atom (Asea-Atom)	Sydkraft
Forsmark 1	BWR/G3	ABB-Atom (Asea-Atom)	ABB-Atom (Asea-Atom)	SSPB
Forsmark 2	BWR/G3	ABB-Atom (Asea-Atom)	ABB-Atom (Asea-Atom)	SSPB
Forsmark 3	BWR/75	ABB-Atom (Asea-Atom)	ABB-Atom (Asea-Atom)	SSPB
Oskarshamn 1	BWR/G1	ABB-Atom (Asea-Atom)	ABB-Atom (Asea-Atom)	OKG
Oskarshamn 2	BWR/G2	ABB-Atom (Asea-Atom)	ABB-Atom (Asea-Atom)	OKG
Oskarshamn 3	BWR/75	ABB-Atom (Asea-Atom)	ABB-Atom (Asea-Atom)	OKG
Ringhals 1	BWR/G1	SSPB/ABB-Atom (Asea-Atom)	ABB-Atom (Asea-Atom)	SSPB
SWITZERLAND (2 units)				
Leibstadt	BWR/6	BBC/Electrowatt	GE	KKL
Mühleberg	BWR/4	BBC/E&B/Getesco	GE	BKW
TAIWAN (4 units)				
Chin-Shan 1	BWR/4	Ebasco	GE	TPC
Chin-Shan 2	BWR/4	Ebasco	GE	TPC
Kuosheng 1	BWR/6	Bechtel	GE	TPC
Kuosheng 2	BWR/6	Bechtel	GE	TPC
USA (37 units)				
Arnold, Duane	BWR/4	Bechtel	GE	IELP
Big Rock Point	BWR/1	Bechtel	GE	CPC
Browns Ferry 1	BWR/4	TVA	GE	TVA
Browns Ferry 2	BWR/4	TVA	GE	TVA
Browns Ferry 3	BWR/4	TVA	GE	TVA
Brunswick 1	BWR/4	UEC	GE	CPL
Brunswick 2	BWR/4	UEC	GE	CPL
Clinton	BWR/6	Sarg&Lun	GE	IPC
Cooper	BWR/4	Burns&Roe	GE	NPPD
Dresden 2	BWR/3	Sarg&Lun	GE	ComEd